

Process Flow Diagram (PFD) for the Flash Column Vaporizer Concentration Section, UOP Phenol Process Unit. The diagram illustrates the flow of cumene, steam, and cooling water through various vessels, pumps, and heat exchangers.

Key Equipment and Streams:

- CONCENTRATION CUMENE QUENCH DRUM D-2301:** Receives cumene from storage (P-4104A/B) and preflash column. It has an overflow to storage tank TK-4104.
- FLASH COLUMN VAPORIZER E-2304:** Receives cumene from the quench drum and preflash column. It has a vent to atmosphere and a reliable cooling water jacket.
- FLASH COLUMN VAPORIZER CONDENSATE PUMPS P-2309A/B:** Pump condensate from the flash column to the quench drum.
- SC3 CONDENSATE:** Condensate from the flash column.
- VENT TO ATMOSPHERE AT SAFE LOCATION:** For excess pressure relief.
- RELIABLE COOLING WATER (NORMALLY NO FLOW):** Jacket water for the flash column vaporizer.
- S3 STEAM:** Steam input to the flash column.
- TI, TIC, FI, LIC, RO:** Temperature, Total Temperature, Flow, Level, and Relief Valve indicators.

Stream Data Summary:

STREAM NUMBER	151	152	320	3241	3251	3411
TEMPERATURE, deg C	38	38	70	97	71	76
PRESSURE, kg/cm2(g)(mm Hg)	----	----	----	----	----	----
TOTAL MASS FLOW, kg/h	3346	3346	85223	97806	97806	12583

Heater and Exchanger Duty Summary:

ITEM NUMBER	E-2304
DUTY, MM kcal/h	4.15

Legend: Dashed line indicates a direct connection between equipment.

Log:

REV.	DESCRIPTION	BY	CHK.	APPR.	DATE
Z1	AS-BUILT	HSO	SWP	JSC	4/25/16
S1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	4/25/14
A1	APPROVED FOR DESIGN	HSO	SWP	JSC	9/12/13
O1	COMMENT ISSUE	HSO	SWP	JSC	7/11/13

Project Information:

- PROJECT NO: 120117
- DWG. NO: 14780-8120-20-23-0002
- SCALE: Z1

HEATER AND EXCHANGER DUTY SUMMARY	
ITEM NUMBER	E-2304
DUTY, MM kcal/h	4.15



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Uop | BASED ON 963766-110-02-A1

PROCESS FLOW DIAGRAM
FLASH COLUMN VAPORIZER
CONCENTRATION SECTION
CONCENTRATION, DECOMPOSITION AND NEUTRALIZATION SECTION
UOP PHENOL PROCESS UNIT

						CONCENTRATION, AND/OR PHENOL CONCENTRATION, AND/OR PHENOL CONCENTRATION						
						UOP PHENOL PROCESS UNIT						
						CERTIFIED FOR CONSTRUCTION			DWG.	DSGN.	CHK.	APPR.
						BY _____		BY				
						DATE _____		DATE				
Z1	AS-BUILT					4/25/16						
S1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	4/25/14	PROJ. NO.		DWG. NO.		REV.	
A1	APPROVED FOR DESIGN	HSO	SWP	JSC	SWL	9/12/13	PEN	120117			Z1	
O1	COMMENT ISSUE	HSO	SWP	JSC	SWL	7/11/13						
REV.	DESCRIPTION	BY	CHK.	APPR.	PROJ. APPR.	DATE	PPCL		14780-8120-20-23-0002			

SCALE